Frozen Foundation-Model Features for ILC vs IDC Classification on TCGA-BRCA

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Abstract

Invasive lobular carcinoma (ILC) and invasive ductal carcinoma (IDC) require different clinical management, but ILC's diffuse, single-cell infiltrative growth pattern makes it harder to classify from histology than IDC's cohesive nests. We evaluate whether *frozen* pathology foundation-model features can reach accuracy comparable to fully fine-tuned, task-specific CNNs on this classification task, using TCGA-BRCA whole-slide images ($n = 109$ patients, 55 ILC / 54 IDC, site-grouped 5-fold cross-validation to prevent scanner/batch leakage). We extract frozen tile features from UNI2-h (1536-dim) and a baseline ImageNet-pretrained ResNet-18 (512-dim), and train 10 classifier-head families on top (logistic regression, SVM, MLP, gated-attention MIL, plain attention, graph attention, and four GAT composites). Our pre-specified primary configuration, UNI2-h with a gated-attention MIL head at default hyperparameters, reaches patient-level AUROC $= 0.917$ (95% CI 0.852, 0.968), comparable to the external benchmark of Ektefaie et al. [2021]: a fully fine-tuned Inception-v3 CNN reporting AUROC $= 0.920$ (95% CI 0.845, 0.993) on the same dataset and task, despite our using zero gradient updates to the backbone. A subsequent 152-point hyperparameter ablation (dropout, optimizer, scheduler, epochs) shows this result is stable, not the product of selecting a favorable configuration: every MIL and plain-attention variant on UNI2-h lands in the 0.90–0.92 range. UNI2-h features outperform the ResNet-18 baseline on every one of 86 matched configurations (median AUROC 0.894 vs. 0.701). We also report three negative results, specified in advance as part of this evaluation's own ground rules: a naive mean-ensemble of correlated classifier heads underperforms its best single member, a zero-shot LLM baseline underperforms a trivial sign-vote control on the same inputs, and SGD optimization consistently fails on the weaker baseline feature space.

1 Introduction

Distinguishing invasive lobular carcinoma (ILC, $\sim$10–15% of invasive breast cancers) from invasive ductal carcinoma (IDC, $\sim$70–80%) from H&E histology is clinically important: the two subtypes differ in growth pattern, metastatic routes, and response to neoadjuvant therapy, but ILC's single-cell "Indian-file" infiltration pattern lacks the cohesive architecture that makes IDC comparatively easy to recognize, and is a frequent source of diagnostic disagreement among pathologists.

Deep learning on whole-slide images (WSIs) has reached strong performance on this task, but historically required full end-to-end fine-tuning of a CNN or transformer backbone on thousands of tiles per slide: expensive in compute, and in curated labeled data. Pathology foundation models (UNI, UNI2-h, CONCH, Virchow, Phikon) pretrained via self-supervision on large, diverse slide corpora offer a cheaper alternative: *frozen* feature extraction followed by training only a lightweight classifier

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head. Whether this frozen-feature approach can reach a *comparable* level of accuracy to task-specific fine-tuning, with a result reported honestly rather than selected from many candidates, is the question this work addresses, with a systematic sweep across classifier architectures and hyperparameters used explicitly as a robustness check on a single pre-specified primary result, not as a search for the highest number.

2 Related Work

Ektefaie et al. [2021], *npj Breast Cancer*, is the directly comparable external benchmark. They fully fine-tuned an ImageNet-pretrained Inception-v3 CNN on the top 200 tiles/slide (ranked by tissue density) from TCGA-BRCA, with patient-level prediction via averaging tile predictions, and splits grouped by tissue source site: the same site-leakage safeguard used in this work. They report patient-level AUC $= 0.920$ (95% CI 0.845, 0.993; tile-level AUC $= 0.800$) for lobular-vs-ductal classification on a held-out TCGA-BRCA test split, from a cohort of $\sim$1,099 TCGA-BRCA patients ($\sim$22% lobular prevalence; held-out test split size for this specific task not stated), and 0.996 on an independent external validation cohort. Our own cohort is a 50/50-balanced subsample rather than this natural prevalence, our interval is computed separately below, and we have not yet evaluated on an external cohort; CPTAC-BRCA is a candidate for future external validation (Section 6). Given the width of both confidence intervals, we describe our result as comparable to theirs, not equivalent to it.

Zhang [2025], arXiv:2508.14779 (ISBI 2026 submission), studies hospital-source bias in frozen UNI/CONCH features and reports disease AUC $= 0.98 \pm 0.01$ for a UNI + MLP baseline on IDC vs ILC, but this number is not a fair comparison target: it is *patch-level* rather than patient-level, computed on only 4,029 patches pre-filtered by CONCH zero-shot to keep high-confidence (easy) patches, from a small cohort (80 WSIs, 4 hospitals) restricted to one demographic group (White female patients aged 60–79). We report it here only for completeness, and separately because the paper's own finding, that hospital/site is strongly predictable from frozen features (AUC $\sim$0.95–0.96), reinforces the importance of the site-grouped cross-validation used throughout this work.

Foundation models. UNI2-h (Mahmood Lab) is a ViT-H/14 vision transformer pretrained via self-supervision on a large multi-site histopathology corpus; we use it here strictly as a frozen feature extractor, with no fine-tuning of backbone weights.

3 Methods

Cohort. 110 TCGA-BRCA patients (55 ILC, 55 IDC) with a usable diagnostic (Primary Tumor, not Metastatic) H&E slide, balanced per tissue-source site; one slide lacked scanner MPP metadata and was excluded (not imputed), leaving $n = 109$ (55 ILC / 54 IDC) across 16 both-class sites.

Tiling and feature extraction. Slides were resampled to a common 0.5 µm/pixel (the smallest target requiring only downsampling across the full cohort), tiled at 224×224px (up to 1,000 tissue tiles/slide by Otsu-threshold tissue fraction), and passed frozen through two backbones: **UNI2-h** (ViT-H/14, 1536-dim) and a **baseline** ImageNet-pretrained ResNet-18 (512-dim, penultimate layer). Both backbones see identical tile coordinates per slide. Extraction throughput was $\sim$5.6–6.6 tiles/s (Apple M1 Pro, MPS backend) for UNI2-h; we did not measure Ektefaie et al.'s fine-tuning cost and so make no direct training-cost comparison.

Cross-validation. Site-grouped 5-fold CV (no tissue-source site split across folds, both classes present in every fold, verified by assertion) repeated across 5 seeds; each patient's final prediction averages its out-of-fold probability across seeds. 95% CIs via patient-level bootstrap (2,000 resamples).

Pre-specified primary configuration. Before any hyperparameter tuning, 10 classifier-head families were each run once at a single default configuration (50 epochs, Adam, lr 3×10^{-4}, dropout 0.25, no scheduler) on both feature sets: (1) logistic regression and (2) SVM (4 kernels) on mean-pooled tile features; (3) a small MLP; (4) gated-attention MIL [Ilse et al., 2018]; (5) plain (ungated) attention pooling; (6) graph attention (GAT, Veličković et al. 2018) over a spatial k-NN tile graph; (7–10) four GAT composites (GAT backbone $\rightarrow$ LR / SVM / MLP / MIL as the final classifier, the LR/SVM variants trained via a differentiable proxy head since those classifiers are not themselves differentiable). UNI2-h + MIL at this default configuration is the primary result reported in Section 4.

Exploratory LLM baseline. Separately, we resolve an open question from this project's own research plan, whether a small LLM can usefully aggregate foundation-model-derived signals without being trained itself, with one concrete, cheap, auditable test: each held-out patient's embedding is reduced to 8 class-oriented signed z-scores on the most train-discriminative feature dimensions, and a small local instruction-tuned LLM (Qwen2.5-0.5B-Instruct, no external API, no cost) is asked to classify from those signals, compared against a trivial no-LLM sign-majority-vote control computed on the identical inputs. We report this as an explicit negative-result test, not a competitive method.

Hyperparameter ablation (robustness check, not model selection). For MLP and MIL heads (both feature sets): dropout $\times\{0.1, 0.2, 0.3\}$ $\times$ epochs $\{50, 100, 150\}$ $\times$ optimizer-profile {Adam@3e-4, SGD@0.01} $\times$ scheduler {cosine, step} $= 36$ combinations $\times\, 4$ base configs $= 144$ runs. SVM kernel ablation: {linear, rbf, poly, sigmoid} $\times\, 2$ feature sets $= 8$ runs. Combined with the 20 default-hyperparameter architecture configs and 2 LLM configs above, 174 total runs were completed. The 2 LLM configs use a different evaluation protocol (Methods, exploratory LLM baseline) and are reported separately; the remaining **172 configurations** (86 per feature set) use the standard CV protocol and are directly comparable to each other and to the primary result.

4 Results

The pre-specified primary configuration, UNI2-h + gated-attention MIL at default hyperparameters, reaches patient-level AUROC $= 0.917$ (95% CI 0.852, 0.968), comparable to Ektefaie et al. [2021]'s fully fine-tuned Inception-v3 benchmark of 0.920 (95% CI 0.845, 0.993), with zero backbone gradient updates. Table 1 reports the subsequent hyperparameter ablation as a robustness check: every one of the top 10 configurations, spanning three dropout values, both optimizers, both schedulers, and three epoch budgets, lands within 0.003 AUROC of each other (0.917–0.920), indicating the primary result is not sensitive to this choice, not that any one tuned variant is a better estimate than the pre-specified one.

Table 1: Top 10 configurations by patient-level AUROC, from the 152-point hyperparameter ablation (robustness check; see Section 4). The pre-specified primary configuration (MIL, base config) is included for reference.

Feature	Head	AUROC	Balanced Acc.	Brier
UNI2-h	MIL (dropout 0.2, Adam, step, 50ep)	0.920	0.817	0.127
UNI2-h	MIL (dropout 0.3, Adam, cosine, 50ep)	0.920	0.826	0.124
UNI2-h	MIL (dropout 0.1, SGD, step, 50ep)	0.919	0.854	0.114
UNI2-h	MIL (dropout 0.1, Adam, cosine, 50ep)	0.919	0.835	0.126
UNI2-h	Attention (plain, default config)	0.919	0.844	0.123
UNI2-h	MIL (dropout 0.3, Adam, step, 50ep)	0.919	0.826	0.126
UNI2-h	MIL (dropout 0.1, Adam, cosine, 150ep)	0.918	0.826	0.132
UNI2-h	MIL (dropout 0.1, Adam, step, 50ep)	0.918	0.826	0.127
UNI2-h	*MIL (pre-specified primary config)*	*0.917*	*0.835*	*0.125*
UNI2-h	MIL (dropout 0.3, Adam, step, 100ep)	0.917	0.826	0.123

Table 2: UNI2-h vs. baseline CNN, across 86 matched configurations per feature set (the full ablation grid plus default-hyperparameter architecture configs, excluding the 2 LLM configs; see Methods).

	UNI2-h	Baseline ResNet-18
Median AUROC	0.894	0.701
Max AUROC	0.920	0.762

UNI2-h's median performance (0.894) exceeds the baseline's maximum (0.762): a complete separation with no overlap across all 172 configurations under the standard protocol.

5 Discussion

MIL and plain attention converge to the same ceiling. Across all dropout/optimizer/scheduler combinations, both gated-attention MIL and plain (ungated) attention independently plateau at 0.917–0.920 on UNI2-h. This suggests the aggregation mechanism's gating component is not the binding constraint on this task: a more sophisticated pooling mechanism is unlikely to help without more training data (Section 6).

Graph attention underperforms plain attention. GAT and its composites reach only 0.88–0.89, below plain attention's 0.919. A naive spatial k-NN graph over tile coordinates does not capture useful structure beyond what unordered attention already extracts: the explicit graph inductive bias is not earning its complexity here.

SGD consistently fails to train on the weaker feature space. All 18 `baseline_cnn + MLP + SGD` ablation configurations reach AUROC 0.43–0.57 with balanced accuracy 0.49–0.55, consistent with a model that failed to leave a poor initialization rather than learning any anti-correlated signal. The identical architecture with Adam reaches 0.73–0.76 (all 18 configurations), and `uni2h + MLP + SGD` trains normally (0.85–0.89, all 18 configurations). We read this as an optimization failure specific to SGD on the weaker, lower-dimensional baseline feature space, not as evidence of a subtler effect; we note it because it recurs across every hyperparameter combination we tried, not because any single run is informative on its own.

A naive ensemble did not help. Averaging one MIL variant's predictions (individually AUROC $= 0.911$; dropout 0.1, Adam, step, 150 epochs, a different hyperparameter setting from the overall-best MIL run in Table 1) with three MLP variants (individually 0.87–0.89) produced a 2-member ensemble AUROC of 0.909 and a 4-member ensemble AUROC of 0.900, both below that MIL run alone. Because all members shared the same UNI2-h feature space, their errors were likely correlated, and mean-averaging a stronger model with weaker ones pulled the average down rather than reducing variance. This motivates cross-foundation-model ensembling (genuinely diverse feature spaces) over same-feature cross-head ensembling as the more promising direction.

The LLM baseline underperforms its own trivial control. The zero-shot LLM classifier reached AUROC $= 0.568$, below the no-LLM sign-vote control (0.670) on the identical input signals. This was run as a deliberately minimal, auditable test of whether a small LLM adds anything over a trivial decision rule on pre-oriented numeric signals; here, it does not. We report it as a clear negative result for this style of LLM-on-tabular-features approach at this scale, not as a claim that no LLM-based approach could work.

6 Limitations and Future Work

Cohort size. $n = 109$ is small for deep learning even with frozen features, and is a balanced subsample of a larger eligible pool (757 IDC + 191 ILC patients with a usable diagnostic slide), not TCGA-BRCA's natural $\sim$22% lobular prevalence. That three architecturally different aggregation mechanisms (gated MIL, plain attention, GAT) all independently converge to the same $\sim$0.92 ceiling is consistent with sample size, not architecture, being the binding constraint. A cohort expansion to **348 patients** (174 ILC + 174 IDC, all available ILC matched per-site with IDC) is in progress at the time of writing, with site-grouped 5-fold feasibility already confirmed.

No external validation. We evaluate only on held-out TCGA-BRCA folds. Ektefaie et al. [2021] additionally validate on an independent cohort (AUC $= 0.996$); we have not yet done so. CPTAC-BRCA is a candidate external cohort for future work.

Calibration. Brier scores are reported on a 50/50-balanced cohort; they do not directly transfer to TCGA-BRCA's natural $\sim$10–15% ILC prevalence, where the same model's calibration would need re-estimating (e.g., via prevalence correction or Platt scaling) before any clinical interpretation. Reliability diagrams are logged per run in our experiment tracker but are not reproduced here.

Single foundation model. Results reported here use only UNI2-h. Phikon-v2 and Virchow2 feature extraction is underway; cross-foundation-model ensembling (motivated by the within-feature-space ensemble's negative result in Section 5) is planned once available.

Frozen features only. We did not evaluate partial or LoRA fine-tuning of the UNI2-h backbone, which carries real overfitting risk at this sample size but could recover additional task-specific signal.

No stain normalization. Cross-site H&E stain and scanner variation was not explicitly corrected for beyond the site-grouped CV safeguard; this is a known confound in multi-institutional cohorts like TCGA.

7 Conclusion

A frozen UNI2-h foundation model with a lightweight gated-attention MIL head, at a single pre-specified default configuration, reaches patient-level AUROC 0.917 (95% CI 0.852, 0.968) on ILC vs IDC classification from TCGA-BRCA whole-slide images, comparable to a fully fine-tuned, task-specific Inception-v3 CNN reporting 0.920 (95% CI 0.845, 0.993) on the same task [Ektefaie et al., 2021], with zero backbone gradient updates. A subsequent 152-point hyperparameter ablation confirms this result is stable rather than a product of selecting among many candidates. Graph-structured attention, naive same-feature ensembling, SGD optimization on weaker features, and a zero-shot LLM baseline each underperformed simpler alternatives; we report these as negative results alongside the positive one, specified as part of this evaluation's own ground rules rather than discovered post hoc. Cohort expansion to 348 patients, external validation, and cross-foundation-model ensembling are the next steps.

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